

COMP47670 Assignment 2 - Spring 2025/26

Deadline: Sunday 26th April 2026

Overview

This assignment involves the analysis of product review data from an online retail website. The dataset contains customer reviews including text content, star ratings, and review helpfulness votes. The objective is to evaluate the performance of different classifiers in the context of two related text classification tasks: (i) predicting review sentiment; (ii) predicting review helpfulness.

The assignment should be implemented as a single Jupyter Notebook (not a script). Your notebook should be clearly documented, using comments and Markdown cells to explain the code and interpret the results of your analysis.

The data for the assignment is available at the link below.

<http://mlg.ucd.ie/modules/python/reviews.zip>

This data contains two related files:

1. *review-text.json*: the text content for each review.
2. *review-scores.json*: the star rating and helpfulness score for each review.

Tasks

1. Data Preparation and Characterisation

- a) Load and parse the data from both input files. Integrate the two data sources by merging them appropriately to create a single dataset. Save the new dataset in an appropriate file format.
- b) Apply any preprocessing steps required to transform the dataset's features into suitable representations for analysis.
- c) Provide a characterisation of the merged dataset, including summary statistics and distributions for the descriptive features.

2. Review Sentiment Classification

- a) Derive a binary sentiment target label ("positive" or "negative") for each review based on its star rating. Create a plot showing the resulting class distribution for the dataset.
- b) Prepare the text data for classification by applying appropriate preprocessing steps to create a numeric representation suitable for machine learning. Then build two different binary classification models using classifiers of your choice to distinguish between "positive" and "negative" reviews.

- c) Evaluate and compare the performance of your two sentiment classification models using an appropriate evaluation strategy. Report your results using suitable metrics and discuss the performance of the models.

3. Review Helpfulness Classification

- a) Derive a binary helpfulness target label ("helpful" or "unhelpful") for each review based on its associated helpfulness information. Create a plot showing the resulting class distribution for the dataset.
- b) Build two different binary classification models using classifiers of your choice to distinguish between "helpful" and "unhelpful" reviews.
- c) Evaluate and compare the performance of your two helpfulness classification models using an appropriate evaluation strategy. Report your results using suitable metrics and discuss the performance of the models. Additionally, compare the overall performance between the sentiment classification task and the helpfulness classification task, discussing any differences you observe.

Use of Generative AI:

- You may use generative AI tools for this assignment (e.g., ChatGPT, Copilot, Gemini), provided you disclose their use as described below.
- At the beginning of your notebook, include an "AI Use Statement" in a Markdown cell. This is **required** even if you do not use any generative AI tools. If you do use generative AI, list the tool (and model, if known) and describe how it was used (e.g., understanding data, debugging or generating code). Penalties will apply if this statement is missing or inaccurate.

Guidelines

- The assignment should be completed **individually**. All submissions will be subject to plagiarism checking. Any evidence of plagiarism can result in a 0 grade.
- The grade awarded will be based on the complexity and appropriateness of the analysis and the level of detail, including data cleaning and preparation, analysis, interpretation, and related aspects.
- See the associated Grading Rubric document for a detailed breakdown of marks for each task.
- Submit your assignment via Brightspace. Your submission should be in the form of a **single ZIP file** which contains:
 - 1. Your Jupyter notebook (IPYNB file).
 - 2. The dataset you saved in Task 1.
- Penalties will apply for late submissions after the specified deadline:
 - 1-5 calendar days late: 1 grade point deduction, e.g. B to B-
 - 6-10 calendar days late: 2 grade point deduction, e.g. B to C+
 - Assignments will not be accepted any later than 10 calendar days without Extenuating Circumstances formally approved by UCD.